Exponential Functions & Growth Models

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Exponential growth and decay basics**

Classify each function as exponential growth or decay, and state the growth/decay factor: $f(x) = 5(1.2)^x$ and $g(x) = 5(0.8)^x$.

$f(x) = 5(1.2)^x$: growth, since the base $1.2 > 1$ (growth rate 20% per unit x). $g(x) = 5(0.8)^x$: decay, since the base $0.8 < 1$ (decay rate 20% per unit x).

2 **[Foundational]** • **Target: Exponential growth and decay basics**

A population of bacteria is modeled by $P(t) = 1000(2)^{t/3}$, where t is in hours. Interpret the meaning of the exponent's structure, and find the population after 9 hours.

The form $2^{t/3}$ means the population doubles every 3 hours. After 9 hours (3 doubling periods): $P(9) = 1000(2)^3 = 1000(8) = 8000$ bacteria.

3 **[Foundational]** • **Target: Visualizing exponential growth**

The graph shows $f(x) = 2^x$ and $g(x) = 3(0.5)^x$, one growing and one decaying.

a Identify which curve represents growth and which represents decay. $f(x) = 2^x$ is the growth curve (increasing, base > 1). $g(x) = 3(0.5)^x$ is the decay curve (decreasing, base < 1).

b State the horizontal asymptote shared by both curves. Both curves approach the horizontal asymptote $y = 0$ (neither ever reaches or crosses it).

(a) $f(x) = 2^x$ is the growth curve (increasing, base > 1). $g(x) = 3(0.5)^x$ is the decay curve (decreasing, base < 1).; (b) Both curves approach the horizontal asymptote $y = 0$ (neither ever reaches or crosses it).

4 **[Foundational]** • **Target: Finding an exponential model from data**

An investment grows from 2000 dollars to 2420 dollars in 2 years under exponential growth $A(t) = A_0(1 + r)^t$. Find the annual growth rate r.

$2420 = 2000(1 + r)^2$: $(1 + r)^2 = 1.21$, so $1 + r = \sqrt{1.21} = 1.1$, giving $r = 0.10 = 10\%$.

5 **[Foundational]** • **Target: Finding an exponential model from data**

A radioactive sample decays from 80 g to 45 g over 6 years. Find the annual decay factor b in the model $A(t) = 80\,b^t$, to four decimal places.

$45 = 80\,b^6$: $b^6 = \frac{45}{80} = 0.5625$. $b = (0.5625)^{1/6} \approx 0.9128$.

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6 [Foundational] • Target: Continuous growth and the number e

A quantity grows continuously according to $A(t) = 500e^{0.03t}$. Find the amount after 20 years, to the nearest whole number.

$A(20) = 500e^{0.03(20)} = 500e^{0.6} \approx 500(1.8221) \approx 911.$

7 [Foundational] • Target: Continuous growth and the number e

Explain the difference between the discrete growth model $A(t) = A_0(1 + r)^t$ and the continuous growth model $A(t) = A_0e^{kt}$, and state how r and k are related when both models describe the same annual growth rate r.

The discrete model compounds growth once per period (e.g. once a year), while the continuous model compounds growth constantly at every instant. They're related by $e^k = 1 + r$, i.e. $k = \ln(1 + r)$ — the continuous rate k is slightly less than the discrete rate r for positive growth, since continuous compounding reaches the same total growth with a slightly lower nominal rate.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Comparing growth rates: exponential vs. polynomial

Explain why, for large enough x, any exponential growth function $f(x) = b^x$ (with $b > 1$) will eventually exceed any polynomial function $g(x) = x^n$, no matter how large the exponent n is. Illustrate with $f(x) = 1.01^x$ vs. $g(x) = x^{10}$.

Exponential functions grow by repeated multiplication, so their rate of growth itself keeps growing exponentially, while a polynomial's growth rate eventually stabilizes into a fixed power-law pattern. Even a very slowly-growing exponential like 1.01^x will eventually overtake x^{10}, because exponential growth compounds indefinitely while polynomial growth does not — it just takes an enormous value of x (in this case, extremely large, since 1.01^x grows very slowly at first) for the crossover to occur.

9 [Intermediate] • Target: Applications: half-life and doubling time

A substance has a half-life of 8 days. Write its decay model $A(t) = A_0\left(\frac{1}{2}\right)^{t/8}$, then find what fraction of the original amount remains after 20 days.

$\frac{A(20)}{A_0} = \left(\frac{1}{2}\right)^{20/8} = \left(\frac{1}{2}\right)^{2.5} = 2^{-2.5} \approx 0.1768$, so about 17.68% remains.

10 [Intermediate] • Target: Applications: half-life and doubling time

A city's population doubles every 15 years and currently stands at 40{,}000. Write an exponential model $P(t)$ and predict the population after 45 years.

$P(t) = 40000(2)^{t/15}$. After 45 years ($t/15 = 3$ doubling periods):
$P(45) = 40000(2)^3 = 40000(8) = 320{,}000.$

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11 [Intermediate] • Target: Transformations of exponential functions

Describe how the graph of $g(x) = 2^{x-3} + 4$ is transformed from the parent function $f(x) = 2^x$, and state the new horizontal asymptote.

$g(x) = 2^{x-3} + 4$ shifts $f(x) = 2^x$ right 3 units and up 4 units. The horizontal asymptote shifts from $y = 0$ to $y = 4$.

12 [Intermediate] • Target: Transformations of exponential functions

Find the horizontal asymptote and y-intercept of $h(x) = -3(2)^x + 6$.

The negative coefficient reflects the graph over the horizontal asymptote line; as $x \to \infty$, $-3(2)^x \to -\infty$, so $h(x) \to -\infty$, while as $x \to -\infty$, $-3(2)^x \to 0$, giving horizontal asymptote $y = 6$. y-intercept: $h(0) = -3(1) + 6 = 3$, so $(0, 3)$.

13 [Intermediate] • Target: Compound interest and financial models

A savings account earns 4% annual interest compounded quarterly, modeled by $A(t) = A_0\left(1 + \frac{0.04}{4}\right)^{4t}$. If 5000 dollars is deposited, find the balance after 3 years, to the nearest dollar.

$A(3) = 5000(1.01)^{12} \approx 5000(1.1268) \approx 5634$ dollars.

14 [Intermediate] • Target: Compound interest and financial models

Explain, without computing exact values, why compounding more frequently (e.g. daily instead of quarterly) at the same nominal annual rate produces a slightly larger balance after the same number of years.

More frequent compounding applies the (smaller) periodic interest rate more often, and interest starts earning interest sooner and more often within each year. This pushes the effective annual growth rate slightly above the nominal rate, so daily compounding yields a marginally larger balance than quarterly compounding at the same stated annual rate.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Finding an exponential model from data • Cross-Lesson: 4.03 Parametric Equations & Modeling

The half-life of Carbon-14 is 5730 years. (a) Write $A(t)$. (b) How much of 200 g remains after 11 460 years? (c) When is only 10% left?

(a) Exponential Decay Function: $A(t) = 200\left(\frac{1}{2}\right)^{t/5730}$. (b) Mass Remaining at $t = 11{,}460$ years: Notice that $\frac{11{,}460}{5730} = 2$ half-lives: $A(11{,}460) = 200\left(\frac{1}{2}\right)^2 = 200\left(\frac{1}{4}\right) = \mathbf{50}$ g. (c) Time to Reach 10% of Initial Mass: Set $\frac{A(t)}{200} = 0.10$: $\left(\frac{1}{2}\right)^{t/5730} = 0.10 \Rightarrow \frac{t}{5730}\ln(0.5) = \ln(0.10) \Rightarrow t = 5730\frac{\ln(0.10)}{\ln(0.5)} \approx \mathbf{19{,}035}$ years.

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16 [Challenging] • Target: Continuous growth and the number e • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

A cup of coffee at 92°C is placed in a room kept at a constant 20°C. After 10 minutes, the coffee has cooled to 68°C. Using Newton's Law of Cooling $T(t) = 20 + 72b^{t/10}$, find the exact temperature after 20 minutes and when the coffee reaches 36°C.

Step 1: Find the 10-minute decay base b: $T(10) = 20 + 72b^1 = 68 \Rightarrow 72b = 48 \Rightarrow b = \frac{48}{72} = \frac{2}{3}$. Step 2: Temperature at $t = 20$ minutes: $T(20) = 20 + 72\left(\frac{2}{3}\right)^{20/10} = 20 + 72\left(\frac{4}{9}\right) = 20 + 32 = \mathbf{52}^\circ$C. Step 3: Solve $T(t) = 36^\circ$C: $20 + 72\left(\frac{2}{3}\right)^{t/10} = 36 \Rightarrow \left(\frac{2}{3}\right)^{t/10} = \frac{16}{72} = \frac{2}{9}$. $t = 10 \cdot \frac{\ln(2/9)}{\ln(2/3)} \approx \mathbf{37.10}$ minutes.

17 [Challenging] • Target: Comparing growth rates: exponential vs. polynomial • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Given that $\cos\alpha = -\frac{5}{13}$ with $\frac{\pi}{2} < \alpha < \pi$, find the exact values of $\sin(2\alpha)$, $\cos(2\alpha)$, and $\tan(2\alpha)$.

Step 1: Find $\sin\alpha$ in Quadrant II ($\sin\alpha > 0$): $\sin\alpha = \sqrt{1 - \left(-\frac{5}{13}\right)^2} = \sqrt{1 - \frac{25}{169}} = \frac{12}{13}$. Step 2: Apply double-angle formulas: $\sin(2\alpha) = 2\sin\alpha\cos\alpha = 2\left(\frac{12}{13}\right)\left(-\frac{5}{13}\right) = \mathbf{-\frac{120}{169}}$.
$\cos(2\alpha) = \cos^2\alpha - \sin^2\alpha = \frac{25}{169} - \frac{144}{169} = \mathbf{-\frac{119}{169}}$. $\tan(2\alpha) = \frac{\sin(2\alpha)}{\cos(2\alpha)} = \mathbf{\frac{120}{119}}$.

18 [Challenging] • Target: Applications: half-life and doubling time • Cross-Lesson: 1.03 Rates of Change & Concavity

Evaluate the exact value of $\sin\left(\arccos\left(-\frac{3}{5}\right)\right)$ and $\tan\left(\arcsin\left(\frac{8}{17}\right)\right)$.

Step 1: Evaluate $\sin\left(\arccos\left(-\frac{3}{5}\right)\right)$: Let $\theta = \arccos(-3/5) \in [0, \pi]$. Since $\cos\theta = -3/5 < 0$, θ is in Quadrant II where $\sin\theta > 0$: $\sin\theta = \sqrt{1 - (-3/5)^2} = \mathbf{\frac{4}{5}}$. Step 2: Evaluate $\tan\left(\arcsin\left(\frac{8}{17}\right)\right)$: Let $\phi = \arcsin(8/17) \in (0, \pi/2)$. Then $\cos\phi = \sqrt{1 - (8/17)^2} = \frac{15}{17}$, so: $\tan\phi = \frac{8/17}{15/17} = \mathbf{\frac{8}{15}}$.

19 [Challenging] • Target: Transformations of exponential functions • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Analyze the polar curve $r(\theta) = 2 - 4\cos\theta$ for $0 \le \theta < 2\pi$. Identify the curve type, find the maximum distance from the pole, and determine the exact angles θ where the curve passes through the pole.

Step 1: Classify the limaçon $r = a - b\cos\theta$: Since $\frac{a}{b} = \frac{2}{4} = \frac{1}{2} < 1$, the curve is a limaçon with an inner loop. Step 2: Maximum distance from the pole: Occurs when $\cos\theta = -1$ ($\theta = \pi$): $r_{\max} = 2 - 4(-1) = \mathbf{6}$. Step 3: Angles at the pole ($r = 0$): $2 - 4\cos\theta = 0 \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \boldsymbol{\theta = \frac{\pi}{3},\ \frac{5\pi}{3}}$.

20 [Challenging] • Target: Compound interest and financial models • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Solve the trigonometric equation $2\cos^2 x - \sin x - 1 = 0$ for all x in the interval $[0, 2\pi)$.

Step 1: Apply the Pythagorean identity $\cos^2 x = 1 - \sin^2 x$:
$2(1 - \sin^2 x) - \sin x - 1 = 0 \Rightarrow 2\sin^2 x + \sin x - 1 = 0$. Step 2: Factor the quadratic in $\sin x$:
$(2\sin x - 1)(\sin x + 1) = 0 \Rightarrow \sin x = \frac{1}{2}$ or $\sin x = -1$. Step 3: Extract all solutions in $[0, 2\pi)$:
$\boldsymbol{x \in \left\{\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}\right\}}$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Exponential growth and decay basics

A population is modeled by $P(t) = 900(3)^{t/3}$, where t is in years. Find $P(6)$.

a. 2700

b. 5400

c. 8100 ✓

d. 24300

Since $t/3 = 2$: $P(6) = 900(3)^2 = 900(9) = 8100$.

22 [Foundational] • Target: Visualizing exponential growth

A population is modeled by $P(t) = 700(3)^{t/3}$, where t is in years. Find $P(3)$.

a. 2800

b. 6300

c. 1400

d. 2100 ✓

Since $t/3 = 1$: $P(3) = 700(3)^1 = 700(3) = 2100$.

23 [Foundational] • Target: Finding an exponential model from data

A population is modeled by $P(t) = 700(2)^{t/2}$, where t is in years. Find $P(4)$.

a. 5600

b. 1400

c. 2100

d. 2800 ✓

Since $t/2 = 2$: $P(4) = 700(2)^2 = 700(4) = 2800$.

24 [Foundational] • Target: Continuous growth and the number e

A population is modeled by $P(t) = 500(3)^{t/3}$, where t is in years. Find $P(3)$.

a. 1000

b. 2500

c. 4500

d. 1500 ✓

Since $t/3 = 1$: $P(3) = 500(3)^1 = 500(3) = 1500$.

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25 [Foundational] • Target: Comparing growth rates: exponential vs. polynomial

A population is modeled by $P(t) = 900(3)^{t/5}$, where t is in years. Find $P(15)$.

a. 26100

b. 8100

c. 24300 ✓

d. 72900

Since $t/5 = 3$: $P(15) = 900(3)^3 = 900(27) = 24300$.

26 [Foundational] • Target: Applications: half-life and doubling time

A population is modeled by $P(t) = 200(2)^{t/5}$, where t is in years. Find $P(20)$.

a. 3200 ✓

b. 6400

c. 1600

d. 3600

Since $t/5 = 4$: $P(20) = 200(2)^4 = 200(16) = 3200$.

27 [Foundational] • Target: Transformations of exponential functions

A population is modeled by $P(t) = 500(2)^{t/5}$, where t is in years. Find $P(20)$.

a. 8500

b. 8000 ✓

c. 4000

d. 16000

Since $t/5 = 4$: $P(20) = 500(2)^4 = 500(16) = 8000$.

28 [Foundational] • Target: Compound interest and financial models

A substance has a half-life of 6 days. What fraction of the original amount remains after 18 days?

a. $\frac{1}{4}$

b. $\frac{1}{16}$

c. $\frac{3}{8}$

d. $\frac{1}{8}$ ✓

Since $t/6 = 3$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^3 = \frac{1}{8}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Exponential Growth and Decay Fundamentals

A substance has a half-life of 5 days. What fraction of the original amount remains after 15 days?

a. $\frac{1}{16}$

b. $\frac{1}{4}$

c. $\frac{1}{8}$ ✓

d. $\frac{3}{8}$

Since $t/5 = 3$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^3 = \frac{1}{8}$.

30 [Intermediate] • Target: Doubling Time & Half-Life Calculations

A substance has a half-life of 6 days. What fraction of the original amount remains after 12 days?

a. $\frac{5}{16}$

b. $\frac{1}{8}$

c. $\frac{1}{4}$ ✓

d. $\frac{1}{2}$

Since $t/6 = 2$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

31 [Intermediate] • Target: Equivalent Exponential Forms & Conversions

A substance has a half-life of 7 days. What fraction of the original amount remains after 21 days?

a. $\frac{1}{8}$ ✓

b. $\frac{3}{8}$

c. $\frac{1}{16}$

d. $\frac{1}{4}$

Since $t/7 = 3$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^3 = \frac{1}{8}$.

32 [Intermediate] • Target: Comparing Linear and Exponential Growth

A substance has a half-life of 4 days. What fraction of the original amount remains after 4 days?

a. $\frac{5}{8}$

b. 1

c. $\frac{1}{4}$

d. $\frac{1}{2}$ ✓

Since $t/4 = 1$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^1 = \frac{1}{2}$.

33 [Intermediate] • Target: Exponential growth and decay basics

A substance has a half-life of 5 days. What fraction of the original amount remains after 25 days?

a. $\frac{5}{32}$

b. $\frac{1}{64}$

c. $\frac{1}{16}$

d. $\frac{1}{32}$ ✓

Since $t/5 = 5$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^5 = \frac{1}{32}$.

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34 [Intermediate] • Target: Visualizing exponential growth

A substance has a half-life of 9 days. What fraction of the original amount remains after 45 days?

a. $\frac{1}{64}$

b. $\frac{1}{16}$

c. $\frac{1}{32}$ ✓

d. $\frac{5}{32}$

Since $t/9 = 5$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^5 = \frac{1}{32}$.

35 [Intermediate] • Target: Finding an exponential model from data

An investment grows from 200 dollars to 128 dollars in 2 years under $A(t) = A_0(1 + r)^t$. Find the annual growth rate r.

a. $-\frac{1}{10}$

b. $-\frac{1}{5}$ ✓

c. $\frac{1}{5}$

d. $\frac{4}{5}$

$128 = 200(1 + r)^2$: $(1 + r)^2 = \frac{128}{200}$, so $1 + r = \frac{4}{5}$, giving $r = -\frac{1}{5}$.

36 [Intermediate] • Target: Continuous growth and the number e

An investment grows from 200 dollars to 162 dollars in 2 years under $A(t) = A_0(1 + r)^t$. Find the annual growth rate r.

a. $\frac{9}{10}$

b. $\frac{1}{10}$

c. 0

d. $-\frac{1}{10}$ ✓

$162 = 200(1 + r)^2$: $(1 + r)^2 = \frac{162}{200}$, so $1 + r = \frac{9}{10}$, giving $r = -\frac{1}{10}$.

37 [Intermediate] • Target: Comparing growth rates: exponential vs. polynomial

Find the horizontal asymptote and y-intercept of $h(x) = 5(2)^x + 1$.

a. Horizontal asymptote $y = 6$; y-intercept (1).

b. **Horizontal asymptote $y = 1$; y-intercept (6).** ✓

c. Horizontal asymptote $y = 1$; y-intercept (5).

d. Horizontal asymptote $y = 0$; y-intercept (6).

As $x \to \pm\infty$ (in the direction $2^x \to 0$), $h(x) \to 1$, giving horizontal asymptote $y = 1$. y-intercept: $h(0) = 5(1) + 1 = 6$, so $(0, 6)$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Applications: half-life and doubling time** • **Cross-Lesson: 1.03 Rates of Change & Concavity**

Find the horizontal asymptote and y-intercept of $h(x) = 3(2)^x - 4$.

a. Horizontal asymptote $y = -1$; y-intercept (-4).

b. **Horizontal asymptote $y = -4$; y-intercept (-1).** ✓

c. Horizontal asymptote $y = -4$; y-intercept (3).

d. Horizontal asymptote $y = 0$; y-intercept (-1).

As $x \to \pm\infty$ (in the direction $2^x \to 0$), $h(x) \to -4$, giving horizontal asymptote $y = -4$. y-intercept: $h(0) = 3(1) - 4 = -1$, so $(0, -1)$.

39 **[Challenging]** • **Target: Transformations of exponential functions** • **Cross-Lesson: 2.02 Logarithmic Functions & Properties**

A savings account earns interest modeled by $A(t) = A_0(1 + \frac{1}{25})^{nt}$, where $nt = 9$. If 5000 dollars is deposited, find the balance to the nearest dollar.

a. 7167

b. 7117 ✓

c. 5000

d. 7067

$A = 5000(1 + \frac{1}{25})^9 \approx 5000(1.4233) \approx 7117$ dollars.

40 **[Challenging]** • **Target: Compound interest and financial models** • **Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle**

A savings account earns interest modeled by $A(t) = A_0(1 + \frac{1}{25})^{nt}$, where $nt = 13$. If 1000 dollars is deposited, find the balance to the nearest dollar.

a. 1000

b. 1515

c. 1665 ✓

d. 1765

$A = 1000(1 + \frac{1}{25})^{13} \approx 1000(1.6651) \approx 1665$ dollars.

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41 [Challenging] • Target: Exponential Growth and Decay Fundamentals • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Which function represents exponential decay with an initial value of 450 and a decay rate of 8% per year?

a. $f(t) = 450(0.92)^t$ ✓

b. $f(t) = 450(1.08)^t$

c. $f(t) = 450(0.08)^t$

d. $f(t) = 450 - 0.08t$

In exponential decay, the base is $b = 1 - r$. With $r = 0.08$, $b = 1 - 0.08 = 0.92$. The model is $f(t) = 450(0.92)^t$.

42 [Challenging] • Target: Exponential Growth and Decay Fundamentals • Cross-Lesson: 4.01 Inverse & Composite Functions

If a quantity grows continuously at an annual rate of 5%, what is the continuous growth model for an initial investment of 1200 dollars?

a. $A(t) = 1200e^{0.05t}$ ✓

b. $A(t) = 1200(1.05)^t$

c. $A(t) = 1200(0.05)^t$

d. $A(t) = 1200e^{1.05t}$

Continuous compounding or continuous growth is modeled by $A(t) = Pe^{rt}$. Here $P = 1200$ and $r = 0.05$, yielding $A(t) = 1200e^{0.05t}$.

43 [Challenging] • Target: Doubling Time & Half-Life Calculations • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

If the doubling time of an investment is 12 years, what is the annual growth factor b in $A(t) = A_0 b^t$?

a. $2^{1/12}$ ✓

b. $12^{1/2}$

c. 2×12

d. $\frac{2}{12}$

Doubling time satisfies $b^{12} = 2$. Taking the 12th root yields $b = 2^{1/12} \approx 1.0595$.

44 [Challenging] • Target: Equivalent Exponential Forms & Conversions • Cross-Lesson: 4.03 Parametric Equations & Modeling

If $P(t) = 250e^{0.08t}$, what is the equivalent discrete annual growth factor rounded to four decimal places?

a. 1.0833 ✓

b. 1.0800

c. 1.0408

d. 1.1664

$e^{0.08t} = (e^{0.08})^t$. Since $e^{0.08} \approx 1.083287$, the annual factor is approximately 1.0833.

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45 [Challenging] • Target: Comparing Linear and Exponential Growth • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

A linear function $L(x) = 50x + 100$ and an exponential function $E(x) = 2^x$ are compared for large $x > 0$. Which statement is true?

a. **Eventually $E(x) > L(x)$ for all sufficiently large x** ✓

b. Eventually $L(x) > E(x)$ because its initial rate of change is 50

c. They will intersect infinitely many times

d. $L(x)$ and $E(x)$ grow at the same rate asymptotically

Any exponential growth function with base $b > 1$ will eventually exceed any linear or polynomial function for sufficiently large values of x.

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